Ibrahim Bukhari

+33744940838 | ibrahimbukhari.ml@gmail.com | Paris, France

Education

Université de Lorraine - ***Joint Masters in Language Communication Technology*** **2025**
- Year 2 of Erasmus Mundus Joint Masters France
- Courses: LLMs, Neural Networks, Computational Morphology, Speech Recognition
- Grade: 15.47 / 20.00

Universidad del Pais Vasco - ***Joint Masters in Language Communication Technology*** **2024**
- Year 1 of Erasmus Mundus Joint Masters Spain
- Courses: Statistics and Mathematics, Machine learning, NLP Application, Deep Learning
- Grade: 8.84 / 10.00

Riphah International University - ***BSc in Computer Science*** **2022**
- Best Final Year project in respective cohort Pakistan
- Grade: 3.66 / 4.00

Skills

Languages: Python, C++, Java, Bash, Slurm
Technology: Linux, Git, Docker, AWS, HPC, Rest API, Tensorflow, Scikit Learn, Huggingface, Langchain, vLLM
AI skills: MlOps, RAG, Agentic Pipelines, Prompt Engineering, Finetuning, Vector Embedding, Semantic Search

Experience

AI Research Intern **Mar 2025 – Aug 2025**
DragonLLM Paris, France
- Contributed to the alignment of financial DragonLLMs to improve reliability and traceability during RAG tasks
- Developed Synthetic Data Generation pipeline using multi-agent setup to produce 40k+ QA samples with exact sentence locations cited with each generated claim in the answer
- Post-trained LLMs using synthetic datasets to add sentence-level citation capability, reducing hallucinations by 17 points and improving citation accuracy by 48 points
- Deployed pipelines on Jeanzay French Supercomputer using Slurm orchestration to utilize multi H100 GPUs
- Created FinBench-Cite benchmark to measure sentence-level citation accuracy and hallucination rate for financial RAG tasks
- **Tech stack**: Python, Huggingface, Transformers, vLLM, FAISS, Slurm

NLP Research Intern **June 2024 – Aug 2024**
Vicomtech San Sebastian, Spain
- Enhanced few-shot entity and intent classification in task-specific dialogue systems by creating synthetic datasets, achieving a 30-point improvement in entity F1 score
- Trained a DistilBERT classifier on seed data to filter mislabeled synthetic examples, reducing mislabeling by 40% resulting in a 3-point increase in overall accuracy
- Generated new synthetic utterances from seed intent–entity pairs using LLMs by substituting entity contexts
- Designed JointBERT-based evaluation to measure intent classification accuracy and entity-level F1 score
- **Tech stack**: Python, DistilBERT, jointBERT, Transformers, tensorflow

Machine Learning Engineer **Apr 2024 – June 2024**
RevelationML Remote
- Designed a RAG-system to query and reason over a proprietary knowledge base
- Created Knowledge Bases by building data ingestion pipelines linked with AWS Bedrock to perform vectorized dataset embedding for efficient retrieval
- Developed Lambda-triggered workflows to rewrite user prompts for optimal information retrieval
- Deployed RESTful APIs enabling seamless integration of the RAG assistant into existing web platforms
- **Tech stack**: AWS (Bedrock, Sagemaker, Lambda, Step-Function, API, Kendra, S3), boto3, Langchain

Projects

Cross-Lingual POS Tagging – Paris, France **Oct 2024 – Feb 2025**

- Evaluated XLM-R and Glot500 models ability for zero-shot cross-lingual transfer in part-of-speech(POS) tagging
- Finetuned using high-resource languages and tested cross-lingual transfer ability on low-resource languages
- Analyzed the effect of subword tokenization on transfer performance
- Achieved up to 72% F1 score on low-resource languages with Glot500 generally outperforming XLM-R
- **Tech stack**: Python, PyTorch, Hugging Face, Transformers, Pandas, NumPy

Multilingual RAG System – San Sebastian, Spain **Apr 2024 – June 2024**

- Architected and deployed a production-ready multilingual RAG system orchestrating 6-stage AI pipeline for biomedical question-answering across languages
- Designed dual-evaluation framework measuring retrieval quality (cosine similarity) and generation accuracy (BLEURT, BLEU, METEOR) against MedQuAD benchmarks
- Developed an agentic RAG workflow that dynamically routes queries through preprocessing, retrieval, and GPT-3.5 generation with custom prompt engineering
- Engineered a multilingual preprocessing pipeline handling language detection, normalization, and bidirectional translation for cross-lingual consistency
- Deployed interactive Streamlit prototype enabling real-time testing and demonstrating full end-to-end workflow from user query to evaluated response
- **Tech stack**: Python, Transformers, LangChain, Chroma DB, SentenceTransformers, Streamlit, OpenAI API

Awards

Erasmus Mundus Scholarship **2023**

- Awarded by the European Commission for Joint Master's in Language and Communication Technologies

Zulfiqar Gold Medal **2022**

- Awarded for achieving outstanding results in BSc Computer Science